

Implementation Verification Report

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This report compares the proposal, presentation slides, database diagram, and actual codebase. The rule used here is conservative: a feature is marked implemented only when there is code, SQL, script, view, or test evidence in the repository.

Source Materials

Source	Location	Use
Project proposal	Provided file: Project Proposal.pdf	Available and text-extracted with PyMuPDF.
Presentation slides	Provided file: Presentation Slides.pptx	Available and slide text extracted from the PPTX package.
Database diagram	Provided file: Database Diagram.jpg	Available and visually inspected.
Actual codebase	Project repository	Inspected from source files, SQL, scripts, views, tests, and existing docs.

Proposed Requirements vs Actual Implementation

Requirement / feature	Source from proposal/slides	Implementation status	Evidence from code files	Notes / recommended action
Offline barangay-office deployment	Proposal pages 2 and 4; slides 10, 13, 34, 36	Implemented	README.md; docs/DEPLOYMENT_GUIDE.md; START-HERE.bat; start-barangay-office.bat	Browser is local interface; backend and database run locally.
Residents coordinate in person, not remotely	Proposal page 2; slides 10, 13	Implemented by scope	No resident self-service public booking routes; auth protects office routes	The prototype/API is for personnel after login.
Monitor available court schedules	Proposal page 3; slide 28; slide 34	Implemented	src/features/schedule/scheduleRoutes.js; views/schedule/index.ejs	Shows available/reserved/missed/cancelled/completed slot states.
Display reserved and available time slots	Proposal page 4; slides 8 and 34	Implemented	scheduleService.js; dashboard.ejs; schedule/index.ejs	Available slots are derived from active time slots minus blocking reservations.
Prevent overlapping schedules	Proposal objectives page 2; expected benefits page 3	Implemented	reservationRepository.js; reservationOverlap.js; database/schema.sql triggers	App-level and database-level conflict protection.
Maximize court utilization	Proposal objective page 2; slide 6	Implemented support	Dashboard counts; nearest available slot; schedule visibility	The system supports utilization through visibility, not automated allocation.
Reservation encoding by barangay personnel	Proposal pages 2-3; slide 10	Implemented	POST /reservations; prototype POST /api/prototype/reservations	Requires authenticated session.
Reservation records administrative interface	Proposal page 4; slide 34	Implemented	views/reservations/index.ejs; reservationRoutes.js	List/filter/export/print/status actions.
Nearest available time if no slot is available today	Proposal page 4; slide 34	Implemented	scheduleService.js findNearestAvailableSlot; dashboardRoutes.js	Dashboard shows nearest available slot in 14-day search window.
Mark missed request as Missed	Proposal page 4; slide 34	Implemented	POST /reservations/:reservationId/status; reservation_statuses seed	Missed is non-blocking after status change.
Admin account creation	Slides 20-26	Implemented	authRoutes.js; userRepository.js; account/create.ejs	Admin-only workflow with required fields and duplicate username validation.
Duplicate username validation	Slide 21	Implemented	DuplicateUsernameError; userValidation.js; users unique index	Both application and database uniqueness exist.
Activity logs	Slide 10; slide 28	Implemented	activity_logs table; activityLogRoutes.js	Reservation actions are recorded; account actions are not logged in current code.

Requirement / feature	Source from proposal/slides	Implementation status	Evidence from code files	Notes / recommended action
ISO 25010 evaluation	Proposal page 2; slide 36	Documented, not an automated test suite	docs/ISO_25010_EVALUATION.md	The repo provides mapping and suggested evaluation tasks.
Formal turnover with documentation and user guides	Slide 36	Implemented documentation; turnover not verifiable	README.md; docs/USER_GUIDE.md; docs/DEPLOYMENT_GUIDE.md	Actual barangay training/sign-off remains an external activity.
Online public booking	Not in scope; slide 13 says offline preferred	Not implemented	No public resident booking routes	Correctly outside current scope.
Online payment	Not proposed	Not implemented	No payment dependencies or routes	Future enhancement only.
Mobile app	Not proposed	Not implemented	No Android/iOS project	Future enhancement only.
Multi-court scheduling	Proposal says the basketball court located in Barangay Sto. Niño	Not implemented	Schema has no courts table	Future enhancement only.

Database Diagram Comparison

Diagram concept	Implemented schema	Status	Comment
STAFF	users	Adapted / implemented	Admin and Staff are unified in users with role and account_status.
password	password_hash	Improved	Implementation stores bcrypt hash instead of plaintext password.
RESIDENTS	residents	Implemented	Fields align with full name, contact number, and address.
RESERVATIONS	reservations	Implemented and expanded	Implementation stores exact start/end time, purpose, remarks, creator/approver, and timestamps.
TIME_SLOTS	time_slots	Implemented	Seeded one-hour active slots from 07:00 to 21:00.
RESERVATION_STATUS	reservation_statuses	Implemented and expanded	Adds status_code, is_blocking, and display_order.
LOGS	activity_logs	Implemented	Stores reservation/user/action/details/timestamp.

Feature Evidence Summary

Feature	Status	Primary evidence
Login and session authentication	Implemented	views/login.ejs / src/features/users/authRoutes.js:27, src/features/users/userRepository.js:17 / users
Role-based access control	Implemented	views/partials/navigation.ejs, views/account/*.ejs / src/features/users/authRoutes.js:204, src/features/users/sessionMiddleware.js:1 / users.role, users.account_status
Admin account creation	Implemented	views/account/index.ejs, views/account/create.ejs, views/account/success.ejs / src/features/users/authRoutes.js:119, src/features/users/userValidation.js:1, src/features/users/userRepository.js:38 / users
Account deactivate/reactivate	Implemented	views/account/index.ejs / src/features/users/authRoutes.js:179, src/features/users/userRepository.js:120 / users.account_status
Change password	Implemented	views/account/password.ejs / src/features/users/authRoutes.js:77, src/features/users/userValidation.js:37, src/features/users/userRepository.js:129 / users.password_hash

Feature	Status	Primary evidence
Dashboard / Home schedule overview	Implemented	views/dashboard.ejs / src/features/schedule/dashboardRoutes.js:18, src/features/schedule/scheduleService.js:86 / time_slots, reservations, reservation_statuses
Daily schedule viewing	Implemented	views/schedule/index.ejs / src/features/schedule/scheduleRoutes.js:12, src/features/schedule/scheduleService.js:11 / time_slots, reservations
Reservation creation	Implemented	views/reservations/new.ejs / src/features/reservations/reservationRoutes.js:182, src/features/reservations/reservationRepository.js:233, src/features/reservations/reservationValidation.js:5 / reservations, residents, reservation_statuses, activity_logs
Reservation overlap prevention	Implemented	Reservation form error display / src/features/reservations/reservationRepository.js:135, src/features/reservations/reservationOverlap.js:41 / database/schema.sql:147, database/schema.sql:183
Reservation editing	Implemented	views/reservations/edit.ejs / src/features/reservations/reservationRoutes.js:90, src/features/reservations/reservationRepository.js:285 / reservations, residents, activity_logs
Reservation status updates	Implemented	views/reservations/index.ejs, views/reservations/show.ejs / src/features/reservations/reservationRoutes.js:214, src/features/reservations/reservationRepository.js:328 / reservation_statuses, reservations, activity_logs
Reservation list, filtering, print, and CSV export	Implemented	views/reservations/index.ejs / src/features/reservations/reservationRoutes.js:29, src/features/reservations/reservationExport.js:13 / reservations, residents, reservation_statuses, users
Activity logs	Implemented	views/activityLogs/index.ejs / src/features/activityLogs/activityLogRoutes.js:13, src/features/activityLogs/activityLogRepository.js:1 / activity_logs
Nearest available slot suggestion	Implemented	views/dashboard.ejs / src/features/schedule/scheduleService.js:66 / time_slots, reservations
Prototype frontend and API bridge	Implemented	public/prototype/sto-nino-court-reservation-system-prototype.html, public/js/prototype-backend.js / src/features/prototype/prototypeRoutes.js:18, src/features/prototype/prototypeApiRoutes.js:36 / users, reservations, residents
Offline startup and maintenance scripts	Implemented	START-HERE.bat, start-barangay-office.bat, maintenance-tools/*.bat / scripts/*.mjs, scripts/*.ps1, src/serverStartup.js / Local MySQL/MariaDB from .env or bundled runtime
Database backup and restore	Implemented as scripts, not in-app buttons	maintenance-tools/backup-database.bat, maintenance-tools/restore-database.bat / scripts/backup-mysql.mjs, scripts/restore-mysql.mjs / MySQL/MariaDB database dump

Open Questions and Verification Gaps

- Final barangay-office deployment sign-off is not provable from the repository alone; it must be rerun on the actual target computer.
- Printer output requires office browser/printer checking.
- Barangay policy rules for approval priority, maximum reservation duration, and allowed reservation days are not encoded beyond date/time conflict prevention.
- Account actions are not currently written to activity_logs; reservation actions are logged.
- There is no in-app backup/restore screen; backup/restore exists as maintenance tooling.